belinus®

AI ENERGY MANAGEMENT SYSTEM · BELINUSAI PLATFORM

Themis

Unlock the power of nature.



NXP EDGE-AI GATEWAY

#EnergyManagementSystem #belinusAI #EdgeAI #Plug&Play #Trading
#Sharing #TariffProtection

01 **belinusAI platform.** Integrates every belinus component into one centrally managed, self-optimising energy system.

02 **Edge-AI hardware by NXP.** On-device neural processing runs site orchestration in real time, cloud-independent for critical decisions.

03 **EU-sovereign AI & servers.** Sovereign European AI models on highly secured EU-hosted servers, with data residency in the Union.

04 **Solar & storage optimisation.** Maximises solar self-consumption and orchestrates battery capacity for superior round-trip efficiency.

05 **Dynamic tariff optimisation.** Plans 24 h ahead against dynamic electricity tariffs, with peak-shaving and negative-pricing protection.

06 **Energy trading & sharing.** Participates in energy markets and shares capacity across the site for maximum value.

07 **EV charger management.** Coordinates ETAPPro charging within the whole-site energy plan.

08 **Always connected.** Your system stays linked to the belinus technical team for monitoring and support.

10 yr	OTA ∞	24/7
HARDWARE WARRANTY	CONTINUOUS PLATFORM UPDATES	BELINUS SUPPORT

Copyright © 2026 Belinus Holding B.V.

www.belinus.com · info@belinus.com

TECHNICAL SPECIFICATIONS

Themis · AI Energy Management System

PLATFORM	EDGE AI	HOSTING	SETUP
belinusAI	NXP NPU	EU Sovereign	Plug & Play

BELINUSAI PLATFORM

Architecture	Hybrid cloud + edge orchestration
Engine	belinusAI predictive optimiser
Provisioning	Plug & Play auto-detection
Optimisation horizon	24 h rolling · real-time
Updates	Continuous OTA

AI DECISION PILLARS

Solar production	Weather-forecast based
Consumption	Household usage patterns
Dynamic tariffs	24 h-ahead pricing
AI settings	User preferences
Site data	Site-specific configuration
Adaptation	Real-time re-planning

EDGE AI HARDWARE

Processor	NXP i.MX application processor
AI accelerator	Integrated NPU, on-device inference
Role	Local orchestration gateway
Resilience	Cloud-independent control
Connectivity	Wi-Fi · Ethernet · 4G
Mounting	Compact wall / DIN gateway

SOVEREIGNTY & HOSTING

AI models	EU sovereign
Servers	EU-hosted, highly secured
Data residency	European Union
Compliance	GDPR
Support link	Always connected to belinus

CAPABILITIES

Solar	Self-consumption optimisation
Storage	Battery capacity management
Tariffs	Dynamic optimisation + protection
Markets	Trading · sharing · flexibility/VPP
Mobility	EV charger management

INTEGRATION

Components	Powerbox · Energywall · Gaia · ETAPPro
Protocols	ModBus · API · OCPP
Monitoring	App & web portal
Scope	Whole-site energy orchestration

Notice: specifications are subject to change without notice.